

Halving goes to the ultraviolet; confinement lives in the infrared

Didactic note, 2026-09-28 (Claude, prompted by the user's question). Confinement and the mass gap are infrared properties, while halving the lattice moves toward the ultraviolet. The refinement programme of the atlas nevertheless bears on the gap, for five reasons, each stated with its source:

1. **One map, two directions.** A halving adds modes at the ultraviolet end; integrating them out again is one Wilson renormalization step toward the infrared. Refinement and coarse-graining are the same pushforward, Proposition 1 of the series/parallel note, read in opposite directions.
2. **The infrared is the invariant of a refinement.** A continuum limit holds the physical scales fixed. In lattice units every infrared quantity shrinks at each halving: am and $a\Lambda$ halve, $a^2\sigma$ quarters. The question a halving asks about the infrared is whether these stay constant in physical units.
3. **The exponential ladder selects what survives.** Along the one-loop trajectory a non-perturbative term $e^{-c/t}$, with $t = \hbar g^2(a)$, equals $(a\Lambda)^{2b_0c}$, so a halving multiplies it by 2^{-2b_0c} . The generated gap, $am \propto e^{-1/(2b_0\hbar g^2)}$, is the term with $2b_0c = 1$: it halves exactly when the lattice halves, and so survives in physical units. Every term with $2b_0c > 1$ dies.
4. **A growing ultraviolet end and a fixed infrared end.** From spacing a to the correlation length ξ there are $\log_2(\xi/a)$ halvings. Refining adds steps only at the ultraviolet end, where t is small; the steps where t is of order one, around $\beta_W \simeq 6$ for $SU(3)$, are the same finite set for every a . Confinement is decided there once; refinement requires the growing number of ultraviolet steps to be controlled uniformly.
5. **Refinement exposes lattice artifacts.** Compact $U(1)$ has a lattice gap that refinement at fixed coupling removes. A property that survives every halving is a property of the continuum theory.

1. One map, two directions

Proposition 1 of the series/parallel note writes the law of the coarse links as the pushforward of the fine heat-kernel measure under the product of half-links. Read from coarse to fine it is a refinement: the fine measure is a consistent extension of the coarse one, with new modes at the shortest scale. Read from fine to coarse it is a blocking step: the mid-plane integral Ψ is the effective action generated by the modes of wavelength between a and $2a$. The same factor Ψ , and the same parallel defect, carry the renormalization in both readings. What the atlas calls “what survives refinement” is, in the other reading, “what is left of the ultraviolet after coarse-graining down to the scale of interest”.

2. The infrared is what a refinement holds fixed

A lattice at spacing a has three scales: a , the correlation length $\xi = \hbar/(mc)$ and the box L . A continuum limit sends $a/\xi \rightarrow 0$ at fixed ξ/L (or with $\xi/L \rightarrow 0$ afterwards). In 1 + 3, one-loop running ties the bare coupling to the spacing through $a\Lambda = e^{-1/(2b_0t)}$, $t = \hbar g^2(a)$, $b_0 = 11N/(48\pi^2)$, so that Λ does not move while a does. The infrared quantities are ratios to Λ : m/Λ , σ/Λ^2 . A halving never computes them; it tests whether they remain what they were. This is the precise sense in which the programme's gap statement (AGENTS.md: which operator, which limits in which order, what the gap is measured in) is a statement about all halvings at once.

In 1+2 the relation is simpler: the heat time $t = \lambda_3 a$ itself halves at every step, the ultraviolet steps are close to Gaussian, and the infrared scale is $\hbar c \lambda_3$ (the cell-7 target $E = C_3 \hbar c \lambda_3$ of the atlas), fixed by the dimensionful coupling.

3. The exponential ladder

At one loop, $e^{-c/t} = (a\Lambda)^{2b_0c}$, and $2b_0c$ is the scaling dimension of the term (atlas §4, “The exponential ladder”). A mass has dimension one, so a physical gap is the term that halves with the lattice: $am \propto (a\Lambda)^1$, the member of the ladder with $2b_0c = 1$. The corrections a single step leaves sit far higher: the bridge image terms at $2b_0c = 11N/3$ (roots) and $11N$ (the $SU(3)$ centre images of a trisection, centre note), beyond the per-volume threshold $2b_0c = 4$ of the zero-spacing note. So every ultraviolet step contains the infrared scale as a tiny non-perturbative term of exactly the size that survives, and nothing else of its kind survives with it. This is dimensional transmutation read step by step. The term is invisible to perturbation theory, since $e^{-1/(2b_0hg^2)}$ has a vanishing Taylor series in g^2 ; it is visible to the exact step, which is why the atlas insists on exact halvings.

4. A growing ultraviolet end and a fixed infrared end

The trajectory from spacing a to the scale ξ has $\log_2(\xi/a)$ halvings. As $a \rightarrow 0$, every new halving is added at the ultraviolet end, where $t = \hbar g^2(a) \rightarrow 0$ and the atlas cells 1–6 give one-step control. The infrared end is the handful of order-one steps where t is of order one; for $SU(3)$ with the Wilson action these sit around $\beta_W \simeq 6$, a few steps above the correlation length (the confinement-scale bands note, with published numbers). That end is the same finite set for every a . Its analysis is the finite verification of the intermediate region; the uniform control of an unbounded number of ultraviolet steps feeding it is what Jaffe and Witten name as the missing idea. The division of labour in the atlas follows: cells 1–6 are the ultraviolet steps, cell 7 is the infrared end.

5. Where confinement can be lost

The compact $U(1)$ theory in $1+2$ is the proved example. Its lattice gap comes from monopoles; at fixed coupling the refinement removes it and the continuum limit is the free photon (Gross), while along the Göpfert–Mack trajectory the gap survives only with a diverging monopole density per physical volume (atlas §4; the monopole note). The ultraviolet step is where a lattice gap is exposed as a lattice artifact. In $SU(3)$ the coupling must run to zero logarithmically (cell 4), and the gap is expected to survive because it is tied to Λ , which the running holds fixed.

6. Newton’s parallel

In Newton’s cell the sagitta and the Galileo action vanish under refinement, $K_\tau \propto \tau^3$ (cut-measure note), while the force survives as the ratio those vanishing quantities define. On the lattice $am \rightarrow 0$ while m/Λ survives. In both cases refinement never computes the surviving quantity directly: it certifies that a ratio stays fixed while every per-cell quantity goes to zero. What refinement cannot remove in either case is the supplied action floor $\hbar$ (atlas §4, “The action floor survives what removes the mass gap”).

7. Consequence for STATE

None; this note explains the programme’s logic for the joint paper and for readers of the atlas.